

Nikhilesh Vijay Chirutala

Software Engineer | Python | AI/ML | AWS

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SUMMARY

Software Engineer with 2.8+ years of experience developing production-grade Python applications, backend services, REST APIs, AI-enabled applications, and cloud-based systems across contact center, CRM, trading, and business platforms. Strong hands-on experience with Python, FastAPI, Django, Flask, AWS, SQL, MongoDB, REST APIs, CI/CD, Git, and production troubleshooting. Experienced in scalable backend development, distributed service integrations, ETL and data processing, cloud storage workflows, automation, monitoring, authentication, authorization, and reliable production deployments across multiple environments. Hands-on experience developing Generative AI applications using RAG and LLM integration for contextual information retrieval and application workflows. Proven ability to design, develop, test, troubleshoot, deploy, and continuously improve secure, reliable, and scalable software components.

TECHNICAL SKILLS

Programming Languages: Python, SQL, JavaScript (ES6+), TypeScript

Software Engineering: Software Development, Object-Oriented Programming, Data Structures & Algorithms, REST API Design, API Development, Backend Development, Microservices, Distributed Systems, Service-to-Service Communication, Technical Troubleshooting, Debugging

AI / ML / GenAI: AI/ML Applications, Generative AI, RAG, Large Language Models (LLM), LLM Integration, Vector Search, Prompt Engineering, Contextual Information Retrieval

Python & Backend: Python, FastAPI, Django, Django REST Framework, Flask, REST APIs, Backend Automation, Business Logic, Data Processing, API Integration

Cloud & AWS: AWS, Amazon S3, Cloud Application Development, Cloud Deployments, Production Cloud Services, Multi-Cluster Deployments, Linux

Databases: MySQL, MongoDB, PostgreSQL, SQL, Database Design

Data Engineering: ETL, Data Processing, Data Validation, Data Recovery, Storage Workflows, Downstream Processing, Data-Driven Applications

DevOps / CI/CD / SDLC: Git, GitHub, GitHub Actions, CI/CD, Software Development Lifecycle (SDLC), Feature Flags, Staged Rollouts, Production Deployment, Automation

Production Reliability: Failure Recovery, Retry Strategies, Idempotent Operations, Validation, Fault Handling, Production Support, Incident Resolution, Root Cause Analysis

Observability: Datadog, StatsD, Sentry, Dashboards, Production Monitoring, Error Tracking, Service Health Monitoring, Application Monitoring

Security: Authentication, Authorization, RBAC, ACL, JWT, Secure API Development, Access Control

Testing & APIs: Unit Testing, API Testing, Postman, REST APIs, Third-Party Integrations, Service-to-Service Integration

Frontend: React.js, Next.js, Angular, TypeScript, Jinja2

WORK EXPERIENCE

Python Software Engineer • CA One Tech Cloud Inc.

Apr 2026 – Present | Bengaluru, India

Client: Convoso — Cloud Contact Center Platform

Python, FastAPI, AWS, Amazon S3, MySQL, Datadog, StatsD, Sentry

- Develop and maintain scalable, production-grade Python and FastAPI services for a high-volume cloud platform, supporting ETL, data processing, service-to-service integrations, storage workflows, automation, monitoring, and production operations.
- Migrated a legacy Perl-based workflow to a long-running Python service deployed across 40 production clusters, implementing feature flags, staged rollouts, validation, rollback, monitoring, and troubleshooting for distributed production workloads.
- Designed and implemented Python integrations with a Call Intelligence microservice, handling service API communication, eligibility checks, account and campaign controls, daily processing limits, retries, response validation, and result persistence.
- Developed reliable data validation and recovery workflows for call-recording processing, handling incomplete jobs, failed operations, inconsistent storage states, and downstream processing dependencies across production environments.
- Supported secure and scalable deployment of Python services across multiple production clusters, coordinating feature-flag configuration, validation, monitoring, troubleshooting, staged rollouts, and production verification.
- Built AWS S3 audit and recovery workflows for recording storage, implementing deterministic object handling and idempotent re-upload operations for incomplete or failed storage migrations.
- Implemented production observability using Datadog, StatsD, dashboards, and Sentry to monitor service health, track application errors, diagnose processing failures, and support production incident resolution.
- Investigated and resolved production issues across Python services, REST APIs, databases, and AWS cloud storage, performing root cause analysis and implementing fixes to maintain reliable and scalable processing workflows.

Software Engineer • Mantra Technologies

Feb 2024 – Mar 2026 | Hyderabad, India

- Worked across four production applications, delivering Python backend services, REST APIs, frontend features, third-party integrations, authentication, and application workflows across the software development lifecycle.

- Designed and developed backend modules, REST APIs, and business logic using Python, FastAPI, and MongoDB for production application workflows and data operations.
- Developed AI-enabled application capabilities, including an AI-powered trading platform and a Retrieval-Augmented Generation (RAG) chatbot for contextual information retrieval using LLM integration.
- Built frontend interfaces using React.js, Next.js, Angular, JavaScript, and TypeScript, integrating application workflows with backend APIs and services.
- Implemented authentication, RBAC, and ACL-based authorization to enforce secure, permission-based access controls across application resources and user workflows.
- Integrated third-party services including Twilio and GitHub API and supported application deployments across AWS and Vercel environments.
- Maintained GitHub Actions CI/CD workflows and contributed to continuous integration, testing, debugging, defect resolution, production support, and reliable application releases.

Key Projects:

AI Bull — Generative AI & RAG Application Platform

Python, FastAPI, RAG, LLM, Vector Search, MongoDB, React.js, Next.js

- Developed an AI-powered trading and investment analysis platform for stock and mutual fund analysis, screening, back-testing, and data-driven application workflows.
- Implemented a Retrieval-Augmented Generation (RAG) based chatbot that retrieves relevant financial and application-specific information to generate contextual LLM-powered responses.
- Integrated Generative AI and LLM capabilities into application workflows, enabling conversational access to relevant financial information and contextual application data.
- Built Python and FastAPI backend APIs and business logic supporting data processing, AI-enabled workflows, and application services.
- Developed back-testing engines and configurable screening and filtering capabilities for stocks and mutual funds.

CRM.io — AI-Driven CRM & Communication Platform

FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, REST APIs

- Developed production CRM modules for Contacts, Tasks, communication workflows, and telephony operations using FastAPI and Python.
- Designed and implemented REST APIs and backend business logic for CRM workflows, application data operations, and service integrations.
- Implemented RBAC and ACL authorization to enforce secure, permission-based access to CRM resources and application workflows.
- Integrated Twilio for telephony and communication functionality and developed server-rendered interfaces using Jinja2.
- Tested APIs and end-to-end workflows using Postman and resolved defects across CRM functionality and production workflows.

GharPe — Real Estate Management Platform

React.js, FastAPI, Python, MongoDB, REST APIs

- Developed the React.js-based administration system for managing real estate projects and controlling their visibility and availability across the platform.
- Built project management workflows allowing administrators to create, update, configure, and manage projects displayed across the platform.
- Developed lead management functionality to track and manage customer inquiries generated through the platform.
- Built builder management features for maintaining builder information and associating builders with their respective projects.
- Developed and integrated FastAPI/Python REST APIs supporting administrative workflows, project configuration, validation, and data operations.

PERSONAL PROJECTS

JS Studio — Online JavaScript Playground

Next.js, GitHub API, Vercel

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application with Next.js and Vercel using a production-oriented application architecture.

AnyScrum — Project Management Platform

Next.js, Neon DB, Clerk, Vercel

- Developed a project management platform supporting projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

Guider — Interactive Learning Portal

Angular 19, TypeScript, SSR, Vercel

- Built an interactive learning platform using Angular 19, TypeScript, and server-side rendering with reusable frontend components.
- Deployed the application on Vercel with a production-ready frontend architecture.

Enterprise REST API Platform — Django and Flask Backend Application

Python, Django, Django REST Framework, Flask, PostgreSQL, REST APIs

- Built modular REST APIs using Django, Django REST Framework, and Flask for authentication, business logic, and data-driven workflows.
- Integrated PostgreSQL with CRUD operations, database models, relationships, validation, and API error handling.
- Implemented JWT authentication and role-based access control with API testing and validation using Postman.

EDUCATION

B.Tech — Electrical & Electronics Engineering

2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur